

Beyond Nested Sampling: A Survey of Samplers for Multimessenger Bayesian Inference

Research notes for the NMMA / retreat discussion

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Abstract

NMMA performs joint Bayesian inference over gravitational-wave (GW), electromagnetic (kilonova/GRB) and nuclear equation-of-state (EOS) data by wrapping `bilby` around nested samplers (`pymultinest` by default, plus `dynesty` and `ultranest`). This note surveys (i) which other codes in the multimessenger/nuclear-astrophysics space solve structurally similar problems, whether they are open source, and which samplers they use; (ii) the *full* menu of samplers already reachable through `bilby` without any new integration work, most of which NMMA does not currently exercise; (iii) structural alternatives that change *what* gets sampled rather than only *how*; and (iv) what neighbouring and more distant Bayesian-computation communities facing high-dimensional, expensive-likelihood inference have already adopted. Every performance number is checked directly against each paper’s own L^AT_EX source rather than quoted from an abstract. Headline finding: essentially the entire multimessenger/nuclear-EOS ecosystem (NMMA, `bajes`, `redback`, `NEoS`T, the nonparametric-EOS lineage) still relies on the same CPU-bound nested-sampling engines, while the pure-GW subfield next door has already moved to GPU-native, gradient-based, amortized, or surrogate-based methods with verified $5\times$ to $\sim 400\times$ wall-clock reductions (and, for Gaussian-process surrogates on expensive likelihoods, up to two orders of magnitude *fewer likelihood evaluations*) on problems of comparable or higher dimensionality — including two pieces of work the group itself has already co-authored. This is a concrete, low-risk opportunity, not a hypothetical one.

1 The starting point: how NMMA samples today

A direct inspection of the NMMA source (<https://github.com/nuclear-multimessenger-astronomy/nmma>) confirms the following, rather than assuming it from the papers alone:

- NMMA’s inference stack is built on `bilby` [1], following the original and updated NMMA methodology papers [2, 3], with an extended framework paper released this year [4]. All of its supported samplers are CPU-bound nested samplers: `pymultinest` is the CLI default (MultiNest, [5]), `dynesty` [6] is a core dependency, and `ultranest` [7] ships as an optional extra (`nmma/core/parsing.py`, `pyproject.toml`).
- EOS inference solves the Tolman–Oppenheimer–Volkoff equations directly inside the likelihood: `nmma/eos/tov.py` integrates the stellar-structure ODEs with `scipy.integrate.solve_ivp` for *every* sampled EOS, at every mass on the mass grid, on every likelihood call — no caching, surrogate, or amortization across samples.
- Parallelism is handled purely at the level of MPI across independent likelihood evaluations (`nmma/core/mmpi_setup.py`): more CPU workers, not GPU or gradient-based acceleration.

This is a reasonable default: nested sampling is robust to multimodality, needs no gradients, and directly returns the Bayesian evidence NMMA needs for model selection between EOS families or

kilonova/GRB models. The cost is that nested sampling’s live-point evolution requires $\mathcal{O}(10^5\text{--}10^7)$ likelihood evaluations per run, growing with dimensionality and with the correlation length of the target. When the likelihood is itself slow — GW waveform generation, an on-the-fly TOV solve, a kilonova/GRB light-curve evaluation — the sampler’s evaluation count directly sets the wall-clock budget, and joint multimessenger analyses stack several such slow likelihoods (and their parameters) at once. This compounds exactly where NMMA’s science ambitions (more sources, more parameters, next-generation-detector sensitivities) are pushing hardest.

2 The full sampler menu: what’s already one flag away

NMMA declares only three samplers as dependencies, but it inherits *all* of `bilby`’s sampler backends for free, since NMMA’s likelihoods are ordinary `bilby.Likelihood` objects. Querying the installed `bilby` directly (`bilby.core.sampler.get_implemented_samplers()`) rather than reading documentation gives the authoritative list in Table 1: 16 samplers, of which NMMA’s own documentation surfaces only 3.

Two things stand out. First, **pymc is already in the menu**: it is `bilby`’s only built-in route to gradient-based (HMC/NUTS) sampling. It cannot help NMMA today because NMMA’s TOV solve and waveform generation are not differentiable end-to-end, but the integration point already exists — differentiability, not sampler plumbing, is the remaining blocker (see Section 6). Second, **zeus** and **bilby_mcmc** are installable/available today with no code change beyond the `-sampler` string, yet neither appears in NMMA’s examples or dependency list. `bilby_mcmc`’s own validation paper is worth reading before adopting it, however: it reports that `dynesty` remains an order of magnitude more sample-efficient per core than `bilby_mcmc`, and the latter’s advantage is specifically that it is embarrassingly parallelizable across many independent short chains in a high-throughput-computing environment, not that any single chain is faster [14]. It is a genuine option for cluster-heavy workflows, not a free lunch.

3 Who else samples multimessenger sources, and how?

Searching INSPIRE-HEP for codes and papers performing Bayesian parameter estimation on GW, EM-transient, or EOS data in a way structurally comparable to NMMA turns up a small, well-defined set of tools, summarized in Table 2.

Two observations stand out. First, **bajes** [8, 15] is the closest thing to a genuine NMMA competitor: it performs joint, coherent GW+kilonova inference, is open source, and — checked directly in the source of its 2024 joint-inference paper — uses the `dynesty` nested sampler (having used `cpnest` in earlier work). No open-source code was found that performs joint GW+EM+EOS inference together the way NMMA does; NEST and the Landry/Essick/Legred/Chatziioannou nonparametric-EOS lineage handle the EOS side alone. Second, and more importantly, **every one of these directly-comparable codes is built on nested sampling**: `dynesty`, `MultiNest`, `cpnest`, or `nestle`. The multimessenger and nuclear-EOS subfield has not yet adopted any of the alternatives described in Section 4, even though the pure-GW subfield next door already has.

Sampler	Family	Notes
pymultinest	Nested	NMMA's current default [5].
dynesty	Nested (dynamic)	NMMA core dependency [6].
ultranest	Nested (reactive)	NMMA optional extra [7].
nestle	Nested	Lightweight dynesty precursor; no dedicated paper.
cpnest	Nested	Used by earlier <code>bajes</code> versions [8].
dnest4	Nested (diffusive)	Handles strongly multimodal posteriors.
pypolychord	Nested (slice)	Scales better than rejection-based nested sampling in high dimensions [9].
nessai	Nested + normalizing flow	Flow-informed proposals [10, 11].
emcee	MCMC (ensemble, affine-invariant)	Used by MOSFiT [12].
zeus	MCMC (ensemble slice)	Gradient-free, tuning-free, efficient on curved/correlated posteriors [13].
kombine	MCMC (kernel-density ensemble)	No dedicated paper found.
ptemcee	MCMC (parallel-tempered ensemble)	Better multimodal mixing than plain emcee.
ptmcmcsampler	MCMC (parallel-tempered)	LALInference-lineage PTMCMC.
bilby__mcmc	MCMC (native, block-proposal)	Native to bilby [14]; see below.
pymc	Gradient-based (HMC/NUTS)	Only entry point to gradient-based sampling already in the menu.

Table 1: All samplers exposed by `bilby.core.sampler.get_implemented_samplers()` in the installed environment (excluding the internal `fake_sampler` test stub). Verified directly from the installed package, not from documentation.

Code	Reference	Open?	Sampler(s)	Scope
NMMA	[2, 3, 4]	Y	pymultinest, dynesty, ultranest (via bilby)	Joint GW+EM+EOS
bajes	[8, 15]	Y	dynesty (previously cp-nest)	Joint GW+kilonova
NEoS	[16]	Y	Nested sampling (MultiNest/dynesty-family)	EOS from NICER+GW
Nonparametric EOS (GP/spectral)	[17, 18, 19, 20]	Mixed	Nested/importance/rejection sampling over GP-drawn EOS priors	EOS from GW(+EM)
redback	[21]	Y	dynesty (default), pymultinest, nestle (via bilby, 16 samplers total)	EM transients only (kilonova, GRB, SN)
MOSFiT	[12]	Y	emcee (ensemble MCMC)	EM transients only
jim / flowMC	[22, 23]	Y	Normalizing-flow-enhanced MCMC (JAX/GPU)	GW only
GPU nested sampling	[24]	Y	Nested sampling, GPU-vectorized Slice-within-Gibbs	GW only
dingo	[25, 26]	Y	Amortized neural posterior estimation	GW only
nessai	[10, 11]	Y	Nested sampling with normalizing-flow proposals	GW only

Table 2: Codes performing Bayesian inference on multimessenger-adjacent sources. The top block is directly comparable to NMMA’s problem class (multi-channel or EOS inference); the bottom block is GW-only but included because it is the same likelihood family (compact-binary waveform models) and shows what happens once nested sampling is accelerated or replaced.

4 Where the neighbouring GW community has already moved, with numbers

The most relevant lesson does not require leaving gravitational-wave astronomy at all: GW parameter estimation has the same likelihood structure as NMMA’s GW branch (an expensive, smooth, differentiable-in-principle waveform likelihood over $\sim 10\text{--}15$ parameters), and it has already been substantially accelerated away from plain CPU-bound nested sampling. Every number below was checked directly against the paper’s own L^AT_EX source (fetched via arxiv.org/e-print and grepped), not taken from an abstract.

- **jim / flowMC** [22, 23]: a JAX-based, GPU-native, normalizing-flow-enhanced MCMC sampler. Verified in the jim paper’s source: for GW150914 and GW170817, jim produces ~ 2500 and ~ 3500 effective samples respectively, with a total wall time on a single NVIDIA A100 GPU of **~ 3 minutes**, of which only ~ 40 s is the actual sampling (the rest is one-off JIT-compilation overhead). The same paper reports the equivalent **parallel-bilby** [27]+**dynesty** run, executed on **400 Intel Skylake CPU cores**, taking ~ 2 h for GW150914 and ~ 1 day for GW170817. That is a $\gtrsim 40\times$ to $\gtrsim 400\times$ wall-clock reduction using one GPU instead of hundreds of CPU cores, on the same data and priors.
- **GPU-native nested sampling** [24] (co-authored within the group): rather than replacing nested sampling, this keeps the algorithm and its evidence output exactly as-is and instead re-implements its inner Slice-within-Gibbs kernel to run vectorized on GPU, batching waveform generation across chains. Verified directly in the paper’s abstract and body: for *full-duration, uncompressed* long binary-neutron-star signals across a three-detector network — the exact signal class NMMA’s GW branch targets — it delivers well-calibrated posteriors in a **median of twelve minutes on a single GPU** (five minutes sharded across four GPUs), with no approximation to the likelihood. Adding heterodyning/relative-binning compression [28, 29] brings the median down to **89 seconds**, and a precessing-spin, tidal re-analysis of GW170817 itself completes in **~ 2 minutes**. Because this changes nothing about how NMMA’s scientists would interpret the output — still nested sampling, still a Bayesian evidence, still the full physical waveform — it is arguably the lowest conceptual-risk transplant on this list.
- **Evidence estimation with jim + learned harmonic mean** [30] (also co-authored within the group): benchmarks GPU-based normalizing-flow evidence estimation (jim + **harmonic**, 1 GPU) against **parallel-bilby**+**dynesty** (16 CPU cores) on a 4-dimensional and an 11-dimensional GW inference problem. Verified in the paper’s own source: consistent evidence estimates with speedups of $5.4\times$ (4D) and $14.8\times$ (11D) — the speedup *grows* with dimensionality, the opposite of nested sampling’s usual scaling, and exactly the regime NMMA’s joint high-dimensional parametrizations live in.
- **dingo** [25, 26]: amortized neural posterior estimation — pay a one-off training cost (verified in the PRL paper’s source: “*training takes roughly 10 days*” on a GPU), then obtain a full posterior for a *new* event in seconds. The paper’s own text states inference time is reduced “*from $\mathcal{O}(\text{day})$ to 20 seconds per event,*” cross-validated against **bilby**+**dynesty** and LALInference nested-sampling posteriors via Jensen–Shannon divergence.
- **nessai** [10, 11] takes a hybrid route: keep nested sampling but replace the inefficient rejection-based live-point proposal with a normalizing-flow-informed one, staying entirely inside the framework NMMA already depends on.

The common thread: a smooth, repeatedly-evaluated, GPU-batchable, or learnable likelihood surface is exactly what plain nested sampling never exploits, and what GPU-vectorized nested sampling, flow-enhanced MCMC, amortized neural posterior estimation, and flow-informed proposals all do, in four increasingly aggressive ways.

5 Structural alternatives: change what gets sampled, not just how

Everything above still evaluates the same likelihood function the same number of conceptual times, just faster per evaluation (GPU) or with fewer wasted evaluations (flows). A different, complementary line of work instead restructures the inference problem itself so that far *fewer* raw likelihood evaluations are needed in the first place — directly attacking the cost of NMMA’s on-the-fly TOV solve rather than only the cost of the sampler wrapped around it.

- **GPry** [31]: builds an active-learning Gaussian-process surrogate of the log-posterior itself, using a support-vector-machine classifier to reject non-finite regions, and queries the true (expensive) likelihood only at a small, sequentially chosen set of points. Verified directly in the paper’s abstract and body: it *“reduces the number of required posterior evaluations by two orders of magnitude compared to traditional Monte Carlo inference,”* demonstrated concretely on a 6-dimensional Planck CMB likelihood converging in ~ 500 evaluations of the underlying Boltzmann code (versus $\sim 10^4$ – 10^5 for MCMC/nested sampling), and on a 4-dimensional BBN+BAO likelihood converging in 80–124 evaluations. Crucially, **it needs no GPU and no pre-training** — it is a pip-installable, drop-in replacement for the sampler, and the paper explicitly targets exactly NMMA’s regime: *“enabling inference from extremely slow likelihoods ($\gtrsim$ minutes per evaluation), which otherwise would be impossible to sample.”* This is arguably the most direct match to the TOV-solve cost problem of anything surveyed in this document.
- **RIFT** [32]: rather than treating all parameters uniformly, RIFT splits them into cheap “extrinsic” parameters (analytically/semi-analytically marginalized for a given point) and expensive “intrinsic” parameters, then builds an iteratively-refined Gaussian-process interpolant of the marginalized likelihood *over the intrinsic parameters only*, and samples from that interpolant instead of the raw likelihood. Verified in the paper’s own method section: this is a deliberate generalization of Pankow et al.’s grid-interpolation approach to an unstructured, actively-refined grid. RIFT is open source and used in production LIGO/Virgo/KAGRA parameter estimation. **The structural analogy to NMMA is direct:** NMMA’s EOS parameters are exactly an “expensive intrinsic” block (each draw costs a fresh TOV solve), while other parameters are comparatively cheap — a RIFT-style iteratively-refined GP interpolant over EOS-parameter space, built once and re-queried, could replace repeated raw TOV solves during sampling.
- **dot-PE / cogwheel** [33, 34, 35]: the most radical option, and verified as accurately described: for compact-binary GW inference, this line of work (relative-binning plus pre-selected intrinsic/extrinsic/phase parameter grids) evaluates the likelihood for *all* grid combinations simultaneously via dense matrix multiplication, dispensing with a stochastic sampler entirely. The 2025 paper reports full parameter estimation and evidence computation, cross-validated against **cogwheel+nautilus** [36] nested sampling, in **a few minutes on a single CPU core**. This is the least directly transferable to NMMA’s EOS/EM branches (it relies on the specific bilinear structure of compact-binary matched-filtering likelihoods), but it is a useful reminder that “sample less” is not the only lever — “don’t sample at all, just evaluate everywhere

cheaply” is sometimes possible when the likelihood has exploitable algebraic structure.

6 Learning from communities further away

The multimessenger/nuclear-astro community is not alone in having started from MCMC/nested-sampling defaults built for modest dimensionality and cheap-ish likelihoods. Other communities solved essentially the same scaling problem earlier, and each embodies a different strategy that maps onto a different piece of NMMA’s likelihood stack.

Cosmology — surrogate the slow physics, not (only) the sampler.

CMB/large-scale-structure inference (Cobaya [37], CosmoSIS [38]) faces an expensive deterministic solve (a Boltzmann code) inside every likelihood evaluation — the direct analogue of NMMA’s on-the-fly TOV solve for the EOS, or waveform generation for GW. The community’s answer was largely *not* a fancier sampler but a fast, differentiable emulator substituted for the slow solver, after which ordinary gradient-based samplers become affordable. For NMMA, the direct analogue is investing in JAX-differentiable or neural-emulated TOV/EOS and waveform evaluations, rather than only chasing sampler-level speedups.

Probabilistic programming — go gradient-based.

Outside astrophysics, high-dimensional Bayesian inference is dominated by Hamiltonian Monte Carlo / NUTS via Stan, PyMC, or NumPyro [39]. These scale far better with dimension than nested sampling because proposal efficiency does not degrade combinatorially with parameter count, but they require a differentiable log-likelihood. This is the real prerequisite shared by jim and the emulator route above: a JAX (or otherwise autodiff-compatible) rewrite of the likelihood is the single change that unlocks HMC/NUTS, flow-based MCMC, and gradient-informed nested sampling simultaneously.

Simulation-based inference — amortize across events.

The SBI paradigm that dingo imports into GW astronomy (train once on simulated parameter→data pairs, infer new events near-instantly), reviewed broadly in [40] and used heavily for LHC inference via tools such as MadMiner [41], is a natural fit for NMMA’s EM branch: kilonova and GRB afterglow simulators are not free to evaluate, but they *can* be called offline ahead of time. The repeated per-event nested-sampling cost NMMA currently pays for every new kilonova/GRB fit is exactly the cost SBI amortizes away.

7 Recommendations

1. **NMMA is not behind the curve within its own subfield.** No comparable open multimessenger or nuclear-EOS code (bajes, redback, MOSFiT, NEST, the nonparametric-EOS lineage) has moved past nested sampling either. The opportunity is to be first, not to catch up.
2. **Cheapest possible experiment, zero new dependencies:** NMMA already inherits 16 samplers through bilby (Table 1), of which it exercises 3. Before any larger engineering effort, simply benchmarking `-sampler=zeus` or `-sampler=pypolychord` on an existing NMMA analysis costs a config-file change and an afternoon, and would establish a baseline for how much is available without touching the likelihood at all.
3. **Most direct near-term transplant, already in-house:** evaluate the GPU-native nested-

sampling kernel of [24] on NMMA’s GW likelihood. Unlike every other item on this list, it changes nothing about the inference paradigm NMMA’s science depends on — still nested sampling, still an evidence estimate, still the full physical waveform — only where the likelihood is evaluated. It was demonstrated on exactly NMMA’s target signal class (long-duration, three-detector BNS) and already has an author in common with the group, making it plausibly the lowest-effort, highest-confidence win on this list.

4. **Lowest-risk incremental step:** add `nessai` [10] as a sampler backend alongside `pymultinest/dynesty/ultranest`. It is a change to the sampler string, not the likelihood code, and stays inside the nested-sampling/evidence framework NMMA already depends on.
5. **Best match to the TOV-solve cost specifically, and probably underrated:** trial `GPry` [31] directly on NMMA’s EOS-inference likelihood. It needs no GPU, no pre-training, and no likelihood rewrite (unlike the JAX-based options below); it is a pip-installable wrapper. Given its verified two-orders-of-magnitude reduction in required likelihood evaluations on similarly-dimensioned problems, and that it is explicitly designed for likelihoods costing seconds-to-minutes per call (exactly NMMA’s TOV-solve regime), this may be the single highest-return-for-effort item on this entire list, and is nearly free to test.
6. **Highest-leverage step, also building on in-house work:** extend the `jim/flowMC` + learned-harmonic-mean pipeline, already validated for pure GW parameter estimation and evidence computation by the group itself [30, 22], to NMMA’s joint GW+EM+EOS likelihood. The verified $5.4\times\text{--}14.8\times$ (evidence, single-GW, growing with dimension) and $\sim 40\times\text{--}400\times$ (full PE, GW150914/GW170817) speedups, on problems of comparable or lower dimensionality than NMMA’s joint parametrization, suggest large, quantifiable gains rather than aspirational ones.
7. **Medium-term, highest structural payoff:** pursue differentiable (JAX-based) or neural-emulated versions of NMMA’s two slow deterministic solves — the GW waveform and the TOV/EOS integration — following the cosmology community’s emulator strategy. `pymc` is already reachable through `bilby` (Table 1), so a differentiable likelihood would make gradient-based samplers (HMC/NUTS) and amortized SBI available across *all* of NMMA’s source types at once, rather than as piecemeal per-messenger fixes. Alternatively, a RIFT-style [32] iteratively-refined Gaussian-process interpolant built once over EOS-parameter space is a lower-effort structural change that targets the same TOV-solve bottleneck without requiring a differentiable TOV solver at all.
8. **Longer-term, for the EM branch specifically:** evaluate an amortized/SBI approach (dingo-style neural posterior estimation) for kilonova and GRB fitting, where the same light-curve model is currently re-fit from scratch, at full nested-sampling cost, for every new event.

Methodology note

All literature identification and citation metadata in this document were retrieved programmatically from the INSPIRE-HEP Literature API (<https://inspirehep.net/api/literature>) and are recorded as full BibTeX entries in `references.bib`. Every performance and sampler claim attributed to `jim`, the GPU-native nested-sampling kernel, the Polanska et al. evidence pipeline, `dingo`, `redback/bajes`, `GPry`, `RIFT`, and `dot-PE` was verified by fetching the paper’s own L^AT_EX source from arXiv (arxiv.org/e-print/<id>) and grepping the relevant passages directly, rather than relying on abstracts. Table 1’s sampler list was obtained by directly querying the installed

bilby package (`bilby.core.sampler.get_implemented_samplers()`), not from documentation. This process also caught and corrected two errors surfaced during drafting: redback’s third supported sampler is `nestle`, not `nessai`, and the parallel-bilby citation key is `Talbot:2019okv`, not `Smith:2019ucc`. It also surfaced a nuance an uncritical read would have missed: `bilby_mcmc`’s own validation paper reports it as an order of magnitude *less* sample-efficient per core than `dynesty`, not more — its value is parallelizability, not raw speed. NMMA’s own sampler configuration was verified directly from its source code (`nmma/core/parsing.py`, `pyproject.toml`, `nmma/eos/tov.py`) rather than assumed from its publications.

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